

Sample

A **sample** is a subset of a population selected for study, used to draw conclusions about that larger population without examining every member of it. Sampling is the practical backbone of statistics: censusing an entire population is often impossible, too expensive, or too slow, so researchers instead collect data from a manageable subset and use it to estimate population characteristics.

The critical requirement is that the sample represent the population well. A sample that systematically differs from the population it's drawn from produces **bias** — a survey of only smartphone users, for instance, cannot fairly represent the opinions of an entire city, since it excludes people without smartphones. The gold standard for avoiding this problem is a **random sample**, in which every member of the population has a known, nonzero chance of being selected, so no subgroup is systematically over- or under-represented.

Worked example. Suppose a university with 20,000 students wants to know the average number of hours students study per week. Surveying all 20,000 students is impractical, so the university selects 500 students using a random number generator applied to the student ID list. The average study time from these 500 students — say, 12.3 hours — becomes an estimate of the true average for the entire student body. Because the sample was chosen randomly rather than, say, by surveying only students in the library, the estimate is far more likely to reflect the whole population accurately.

Problem-solving application. When designing or evaluating a sample, ask three questions: (1) Does every member of the population have a chance of being included? (2) Is the sample large enough to reduce the effect of random chance (larger samples generally yield more stable estimates)? (3) Is there a hidden factor — timing, location, method of contact — that could exclude or overrepresent certain groups? For example, a phone survey conducted only during weekday work hours will oversample retirees and undersample working adults, biasing results even though selection appears random within the pool of people who answer. Recognizing these pitfalls is essential before trusting any conclusion drawn from

sample data, since the validity of statistical inference depends entirely on how well the sample mirrors the population it claims to describe.

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